Example 3.8 i) Add: $3x - 4y + z$ and $2x - z + 3y$ ii) Subtract $2x - 5y$ from $4x + 3y$

Solution

i) $(3x - 4y + z) + (2x - z + 3y)$

$= (3x + 2x) + (-4y + 3y) + (z - z)$

$= (3 + 2)x + (-4 + 3)y + (1 - 1)z$

$= 5x - 1y + 0z$

$= 5x - y.$

Aliter :

	$3x$	$-4y$	$+z$
(+)	$2x$	$+3y$	$-z$
	$5x$	$-y$	$+0$

ii) $(4x + 3y) - (2x - 5y)$

$= (4x + 3y) + (-2x + 5y)$

$= (4x - 2x) + (3y + 5y)$

$= (4 - 2)x + (3 + 5)y$

$= 2x + 8y.$

Aliter :

	$4x$	$+3y$
(–)	$+2x$	$-5y$
	(–)	(+)
	$+2x$	$+8y$

Think

What will you get if twice a number is subtracted from thrice the same number?

Example 3.9

Mani and his friend Mohamed went to a hotel for dinner. Mani had 2 idlies and 2 dosas whereas Mohamed had 4 idlies and 1 dosa. If the price of each idly and dosa is x and y respectively, then find the bill amount in x and y.

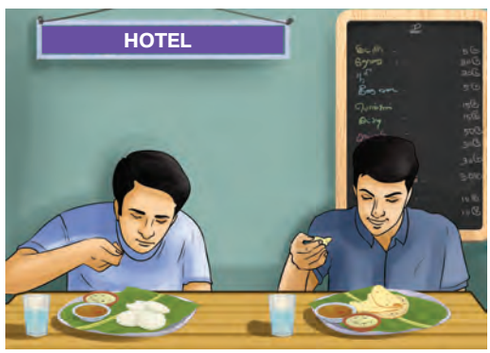


Fig.3.6

Solution

Given that, the price of one idly is 'x' rupees and the price of one dosa is 'y' rupees

So, Mani's bill amount: $(2 \times x) + (2 \times y) = (2x + 2y)$

Mohamed's bill amount: $(4 \times x) + (1 \times y) = 4x + y$

Therefore, the total bill amount $= (2x + 2y) + (4x + y)$

$= (2 + 4)x + (2 + 1)y = 6x + 3y.$

Aliter:

In total, they had $2 + 4 = 6$ idlies. i.e. $6 \times x = 6x$

and $2 + 1 = 3$ dosas. i.e. $3 \times y = 3y$

Therefore, the total bill amount $= 6x + 3y.$

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